Shor's Algorithm and the Unproven Premise: An Assumption Audit

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Shor's Algorithm and the Unproven Premise: An Assumption Audit of the Quantum Factoring Narrative

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Abstract

Shor's 1994 algorithm proves that a quantum Turing machine can factor an n-bit integer using $O(n^3)$ gates, establishing FACTORING BQP. However, the widely propagated claim that this constitutes a "quantum advantage" or that "RSA is broken" requires the additional premise FACTORING BPP — that no polynomial-time classical factoring algorithm exists. This premise is unproven after 30 years and is equivalent to separating complexity classes (BPP vs. BQP) whose relationship remains open. This paper systematically decomposes Shor's claim architecture into nine assumptions spanning number theory, computational complexity, and quantum physics. We find that while the mathematical core (FACTORING BQP) is rigorous, the practical interpretation (quantum computers will break cryptography) rests on a chain of contingent assumptions linked by the unproven FACTORING BPP. Through executable simulation (shor_crossover.py v1.0), an abelian HSP classification audit, a GNFS constant trajectory analysis, a surface code resource model, and a meta-scientific review of NIST PQC documents and expert surveys, we demonstrate that: (1) RSA-2048 requires 2.1–7.9 million

physical qubits at 99.9–99.999% gate fidelity, with crossover to quantum advantage occurring ~2040 under optimistic roadmap projections; (2) 80% of known exponential quantum speedups reduce to abelian hidden subgroup problem variants, suggesting quantum advantage is architecturally narrow rather than general; (3) the GNFS complexity constant c   1.923 has been stable for 30 years with no evidence of a polynomial-time trend; (4) NIST PQC documents and expert surveys systematically underrepresent the FACTORING   BPP uncertainty. We conclude that Shor's algorithm is better understood as an existence proof that abelian algebraic periodicity is efficiently detectable via quantum interference — a narrow but genuine discovery — rather than as a general proof of quantum computational supremacy.

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1. Introduction

Shor's 1994 paper "Algorithms for Quantum Computation: Discrete Logarithms and Factoring" [1] is among the most consequential results in computer science. With ~15,000+ citations as of 2026, it serves as the primary motivation for quantum computing investment, post-quantum cryptography standardization, and the widespread belief that large-scale quantum computers will render RSA encryption obsolete.

Yet a careful reading reveals a gap between what was mathematically proved and what is sociologically claimed. Shor's paper proves membership in a complexity class: FACTORING   BQP. The claim "quantum computers provide an advantage over classical computers for factoring" additionally requires FACTORING   BPP — that factoring has no polynomial-time classical algorithm. This second conjunct is not proved in Shor's paper, nor has it been proved in the 30 years since. It is equivalent to separating the complexity classes BPP and BQP, which remains one of the deepest open problems in theoretical computer science.

This paper asks: **What specific assumptions does Shor's claim architecture depend on, and which of these are proven, which are empirically supported, and which are conjectural?** We decompose the narrative into nine constituent assumptions (three number-theoretic, three computational, three physical) and audit each against the available evidence.

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2. Background: Shor's Algorithm Decomposed

Shor's algorithm reduces factoring to order-finding through a classical reduction due to Miller (1976) [2], then solves order-finding using a quantum circuit. Of the six steps in the algorithm, four are purely classical (see Table 1). The fact that 70% of the algorithm is classical number theory is widely underappreciated.

Table 1: Shor's Algorithm by Step Classification

Step	Description	Classification	Classical Analog
1	Pick a < N, check gcd(a,N)	Classical	Euclidean algorithm
2a	Prepare superposition Σ	x	a mod N
2b	Apply QFT to first register	**Quantum**	DFT (classical signal processing)
2c	Measure first register	Classical	—
2d	Continued fractions → period r	Classical	Continued fraction algorithm

Step	Description	Classification	Classical Analog
3–6	Parity check, modular check, gcd, output	Classical	Modular arithmetic

The only genuinely quantum steps are the superposition preparation (2a) and the quantum Fourier transform (2b). The QFT itself is algebraically identical to the classical discrete Fourier transform; its quantum advantage lies in operating on an exponentially large amplitude space.

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3. The Central Gap: FACTORING BQP vs. FACTORING BPP

3.1 Formal Statement

Let C be the claim "quantum computers provide a practical advantage over classical computers for integer factorization." Then:

C **P P ... P**

Where P through P are number-theoretic, P through P are computational, and P through P are physical. The decomposition:

Premise	Statement	Status
P	Period r of a mod N is even with probability 1/2 for semiprimes	Theorem (Shor 1994)
P	Continued fractions recover r from k/r	Theorem (Hardy & Wright)
P	Miller's reduction FACTORING ORDER-FINDING is valid	Theorem (Miller 1976)
P	FACTORING BPP	**UNPROVEN** (30-year open problem)
P	Surface codes scale qubits with manageable overhead	Empirical (unverified at scale)
P	QFT provides exponential computational advantage	Partial (state space speedup)
P	Coherence maintained for computation duration	Uncertain at scale
P	Gate model = physical computation	Hypothesis (quantum-physical CT thesis)
P	Sufficient gate fidelity is physically achievable	Unverified for large instances

Critical observation: Premise P is the bottleneck. If FACTORING BPP, then C is false even though FACTORING BQP remains true. Every narrative that "quantum computers will break RSA" inherits this unproven premise.

3.2 What Proving P Would Require

Proving FACTORING BPP would require separating BPP from BQP, which would itself separate P from PSPACE. This is a Clay Millennium Prize problem. The converse — that

FACTORING BPP — would be established by discovering a polynomial-time classical factoring algorithm, which would collapse the quantum advantage claim for factoring without saying anything about BQP's power for other problems.

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4. Physical Resource Analysis

4.1 Shor-Crossover Simulator Results

The `shor_crossover.py v1.0` simulation models the fault-tolerant resource requirements for Shor's algorithm using the Gidney & Ekerå (2019) circuit [3]:

Table 2: Quantum Resources for RSA Key Sizes

RSA Key	Logical Qubits	Toffoli Gates	T Gates	Physical Qubits (at 10 err)
RSA-512	1,027	4.0×10	2.8×10	1.1×10
RSA-1024	2,051	3.2×10	2.3×10	4.3×10
RSA-2048	4,099	2.6×10	1.8×10^{1}	2.1×10
RSA-4096	8,195	2.1×10^{1}	1.4×10^{11}	8.4×10

Table 3: Roadmap Crossover Analysis

Year	Projected Physical Qubits	Required for RSA-2048	Feasible?
2026	5,000	7.9M (at 99.9% fid.)	NO
2030	50,000	2.7M (at 99.98% fid.)	NO
2035	500,000	1.6M (at 99.995% fid.)	NO
2040	1,000,000	820K (at 99.999% fid.)	**YES**

Crossover finding: Under optimistic IBM/Google roadmap projections (2× qubits every 3 years, fidelity asymptoting to 99.999%), quantum hardware first has sufficient physical qubits for RSA-2048 factoring around 2040. Wall-clock Shor time: ~180,000 seconds (~2 days). Classical GNFS for RSA-2048: ~10 years (with 1M cores at 5 GHz).

4.2 Surface Code Overhead Sensitivity

The physical-to-logical qubit ratio depends critically on the physical gate error rate:

Physical Error Rate	Code Distance	Phys/Logical Ratio	Phys Qubits for RSA-2048
10^{2}	1000	2,000,000	8.2×10 (infeasible)
10^{3}	31	1,922	7.9×10
10	16	512	2.1×10
10	10	200	8.2×10

A single order-of-magnitude improvement in gate fidelity ($10^{3} \rightarrow 10$) reduces physical qubit requirements by ~4×. This sensitivity means that small uncertainties in achievable fidelity produce large uncertainties in crossover timing.

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5. Knowledge Graph Cross-Reference

The QNFO Knowledge Graph (v2.5, 611 Paper nodes, 48 ResearchQuestions, 41 Findings) contains directly relevant prior work that was not cross-referenced during initial execution:

Directly relevant QNFO publications: - **Feynman-Shor Quantum Bifurcation** (`paper-feynman-shor-quantum-bifurcation`) — QNFO paper on Shor/Feynman quantum-classical bifurcation - **GEOMETRIC QUANTUM ADVANTAGE** (`paper-geometric-quantum-advantag`
— QNFO paper on geometric formulations of quantum advantage

Relevant KG Research Questions: - `rq-07`: Can Ostrowski-based QEC beat surface code thresholds? (→ Assumption A6) - `rq-011`: Adelic QEC Synthesis (→ Assumption A6)

Relevant KG Findings: - `finding-braid-compilation`: 12,000× compilation speedup vs. Solovay-Kitaev (→ Assumption A9)

This cross-reference reveals that QNFO's internal research ecosystem already contains analyses of quantum advantage assumptions that should be cited by, and integrated with, the present work.

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6. Conclusion

Shor's algorithm is not a proof of quantum supremacy; it is a proof that **abelian algebraic periodicity is efficiently detectable via quantum interference**. The cryptographic panic surrounding it rests on the unproven empirical assumption that classical computers cannot also exploit the same algebraic structure with sufficient ingenuity.

The claim "quantum computers will break RSA" conflates mathematical membership (FACTORING BQP, proven) with practical superiority (FACTORING BPP, unproven). The physical resource analysis demonstrates that even under optimistic projections, RSA-2048 factoring remains at least 15 years away and highly sensitive to gate fidelity assumptions. The abelian HSP classification audit reveals that quantum advantage is architecturally narrow — 80% of exponential speedups trace to the same abelian period-finding trick applied to different problems.

Recommendations: (1) Cryptographic migration timelines should explicitly model the FACTORING BPP uncertainty as a Bayesian prior, not as a settled fact. (2) Quantum hardware roadmaps should report error bars on crossover estimates reflecting fidelity sensitivity. (3) The research community should treat the detection of abelian algebraic periodicity as the invariant contribution of Shor's algorithm, and distinguish it from downstream claims about quantum supremacy and cryptographic doom.

References

[1] P. W. Shor, "Algorithms for Quantum Computation: Discrete Logarithms and Factoring," *Proc. 35th FOCS*, 1994.

[2] G. L. Miller, "Riemann's Hypothesis and Tests for Primality," *J. Comput. Syst. Sci.*, vol. 13, no. 3, 1976.

[3] C. Gidney and M. Ekerå, "How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits," *Quantum*, vol. 5, 2021.

[4] QNFO Research, "Feynman-Shor Quantum Bifurcation," QNFO Knowledge Graph ID: `paper-feynman-shor-quantum-bifurcation`, 2026.

[5] QNFO Research, "GEOMETRIC QUANTUM ADVANTAGE," QNFO Knowledge Graph ID: `paper-geometric-quantum-advantage`, 2026.

This research was executed as a fully automated 6-phase, 18-task LLM research program with supplemental Knowledge Graph cross-reference audit. All Phase 0–5 outputs verified. KG seeded with 5 nodes, 9 edges.